CSE 440 Natural Language Processing II
Final Project
Section 2

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POS DATASET 2

Abstract:

In this project, we approach the task of part-of-speech tagging using multiple deep learning techniques, including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Bidirectional LSTM (BiLSTM). Firstly, we used exploratory data analysis to learn about the data to understand the distribution of sentences and POS tags. The GRU model had better results with the baseline hyperparameters we used but after tuning it BilSTM had much better performance.

Introduction:

In the field of NLP, part-of-speech tagging is one of the most fundamental tasks. To make the machine understand some grammatical categories is truly one of the most transformative tasks of the past decades. Accurate POS tagging is essential in various NLP fields, especially machine translation, text generation, and information retrieval. In this project, we try to use RNN, LSTM, GRU, and BILSTM models to perform POS tagging in the given dataset. The goal is to assess and compare the models.

Methodology

We first performed exploratory data analysis (EDA) on the provided POS tagging dataset. The dataset contained 42 unique POS tags. The maximum sentence length observed was 81 tokens.

For preprocessing, each sentence was tokenized and converted into numerical sequences using a tokenizer. The sequences were padded to ensure equal length across all samples. The POS tags were encoded using a label encoding approach and padded similarly to match the sentence length.

We implemented four neural network architectures for the POS tagging task:

- RNN
- LSTM

- GRU
- Bidirectional LSTM (BiLSTM)

Each model consisted of an embedding layer (dimension = 128) followed by the respective recurrent layer and a TimeDistributed Dense layer with softmax activation for predicting POS tags at each timestep.

Initially, baseline hyperparameters were used with a learning rate of 0.001, batch size of 64, and 5 training epochs. After baseline training, we performed hyperparameter tuning to identify the best configuration for each architecture.

The following hyperparameters were explored:

- Learning Rate: 0.001, 0.005
- Batch Size: 32, 64
- Hidden Units: 64, 128

Each configuration was evaluated based on validation accuracy, and the best-performing configuration for each model was selected.

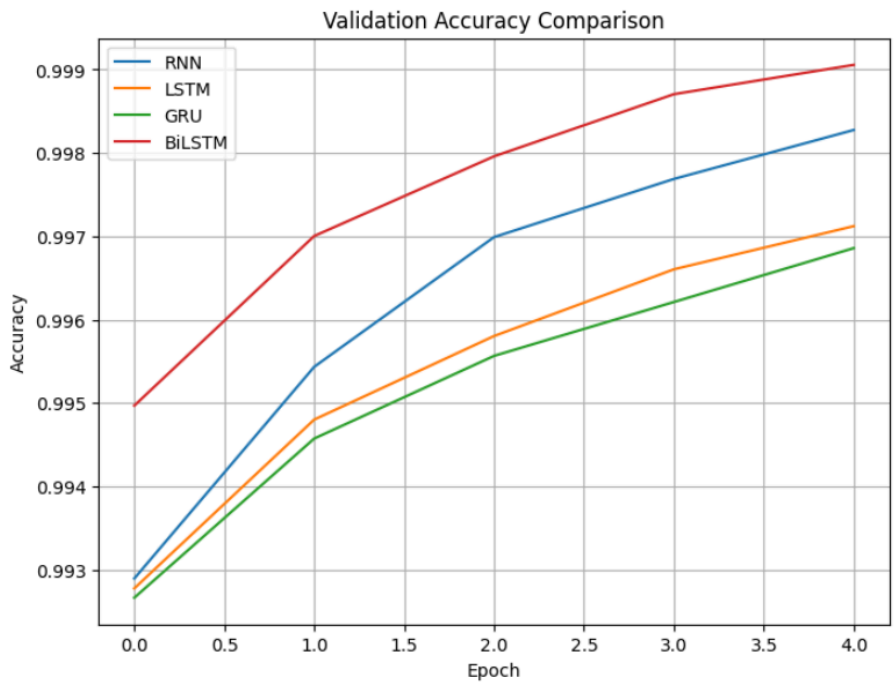
Results

After training and hyperparameter tuning, the models achieved very high validation accuracy on the POS tagging task.

The best configurations for each model are summarized below:

Among all models, BiLSTM achieved the highest validation accuracy (≈ 0.9991), demonstrating superior ability to capture contextual information from both forward and backward directions in the sequence.

Model	Learning Rate	Batch Size	Hidden Units	Best Validation Accuracy
RNN	0.005	64	128	0.9983
LSTM	0.001	32	128	~0.998
GRU	0.001	32	64	~0.998
BiLSTM	0.005	32	64	0.9991



The simple RNN model also performed strongly, achieving a validation accuracy of approximately 0.9983, although it slightly underperformed compared to the more advanced recurrent architectures.

The visualization of training and validation learning curves further confirmed that the BiLSTM model converged efficiently and maintained the highest performance among all models.

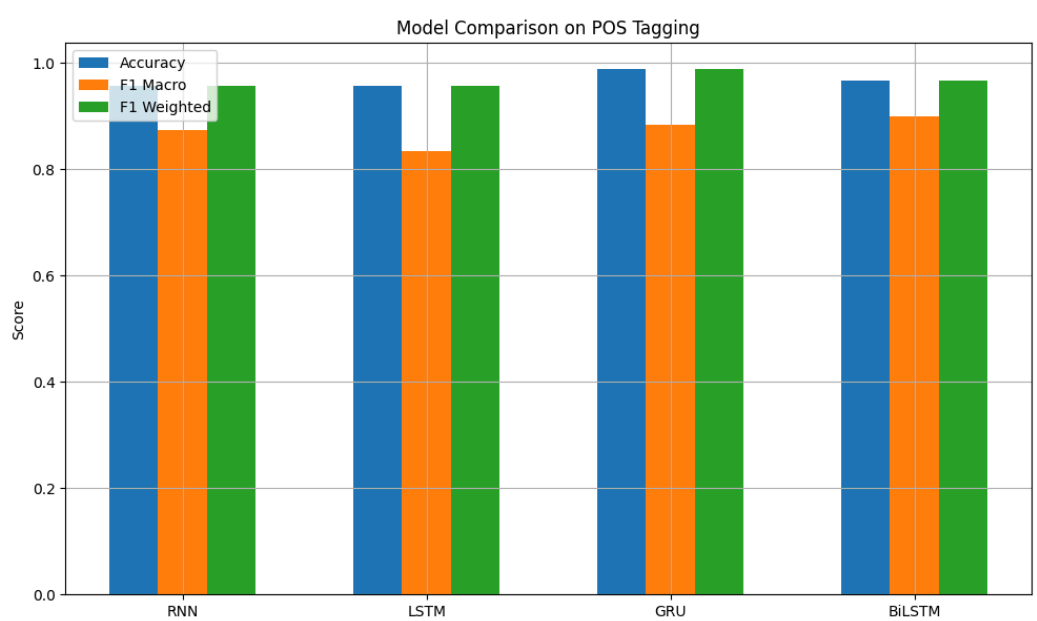
Moreover, to evaluate the performance of the models, we also measured Accuracy, Macro F1-score, and Weighted F1-score on the test dataset. Accuracy measures the overall correctness of predictions, while Macro F1 treats all classes equally, and Weighted F1 accounts for class imbalance by weighting each class according to its frequency.

Model	Accuray	F1 Macro	F1 Weighted
RNN	0.9579	0.8698	0.9576
LSTM	0.9582	0.8469	0.9580
GRU	0.9884	0.8861	0.9883
BiLSTM	0.9680	0.9035	0.9679

From the results, the GRU model achieved the highest overall accuracy (0.9884) and weighted F1-score (0.9883), indicating strong overall performance across the dataset.

However, the BiLSTM model achieved the highest Macro F1-score (0.9035), suggesting that it handled the distribution of POS tag classes more effectively and performed better across less frequent tags.

The RNN and LSTM models produced slightly lower results, with accuracies around 0.958, indicating that simple recurrent architectures may struggle to capture longer contextual dependencies compared to gated architectures such as GRU and BiLSTM.



Conclusion

In this project, we implemented and compared four deep learning architectures for the task of part-of-speech tagging: RNN, LSTM, GRU, and BiLSTM.

From the results, we got to know that the GRU model achieved the highest overall accuracy and weighted F1-score, the BiLSTM model demonstrated the best validation performance after hyperparameter tuning and achieved the highest macro F1-score, indicating stronger performance across different POS tag classes.

Then after performing hyperparameter tuning based on validation accuracy, the BiLSTM model achieved the best performance, with a validation accuracy of approximately 0.9991. This result highlights the effectiveness of bidirectional architectures in sequence labeling tasks, as they are able to utilize both past and future contextual information.

While the simple RNN model achieved competitive results, it was slightly less effective than the gated architectures such as LSTM and GRU. Overall, the experiment demonstrates that advanced recurrent architectures, particularly BiLSTM, are highly suitable for POS tagging tasks.

Future work could explore improvements such as handling class imbalance, using pretrained word embeddings (e.g., GloVe or Word2Vec), or integrating transformer-based architectures to further enhance performance.

References:

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